

# Sentinel IDS: A Predictive Intrusion Detection System Using Temporal Machine Learning and Explainable Artificial Intelligence

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**Undergraduate Research Paper (UGRF)**

## **Abstract**

Intrusion Detection Systems (IDS) remain one of the most important components of modern cyber defense architectures. However, most existing IDS solutions operate reactively by identifying attacks after malicious activity becomes observable. Sentinel IDS proposes a predictive security framework that investigates whether temporal patterns extracted from network traffic can be used to forecast cyber attacks before execution. The framework combines machine learning, deep learning, network traffic analytics, threat intelligence, and Explainable Artificial Intelligence (XAI) within a unified platform. Sentinel IDS introduces a prediction horizon concept that evaluates how early an attack can be anticipated. The platform includes real-time monitoring, attack forecasting, SHAP-based explanations, and research analytics dashboards. Experimental evaluation is currently ongoing; therefore, no performance metrics are reported in this paper.

## **Keywords**

Predictive Security, Intrusion Detection Systems, Cybersecurity, Explainable AI, Machine Learning, Deep Learning, Threat Intelligence, Attack Forecasting, XGBoost, LSTM, GRU

## **1. Introduction**

The increasing complexity of cyber attacks has challenged conventional defense mechanisms. Modern organizations face ransomware campaigns, advanced persistent threats, botnets, insider threats, and multi-stage attacks that evolve continuously. Traditional IDS technologies focus primarily on detecting attacks after indicators of compromise become visible. While these systems remain valuable, they often provide limited time for response and mitigation.

The vision of Sentinel IDS is to move from reactive security toward proactive security. Instead of asking whether an attack is currently occurring, Sentinel IDS investigates whether a future attack can be forecast using historical and temporal behavioral patterns. Such a capability could provide security analysts with early-warning intelligence and additional response time.

## **2. Problem Statement**

Current intrusion detection systems are fundamentally reactive. Security teams often receive alerts only after malicious actions have started. This research addresses the following question:

Can machine learning and temporal behavioral analysis predict cyber attacks before execution and provide actionable early-warning intelligence?

Answering this question requires investigating temporal relationships within network traffic, identifying behavioral precursors, and evaluating predictive models capable of forecasting future malicious activity.

## **3. Research Objectives**

Primary Objective:

Develop a predictive intrusion detection framework capable of forecasting cyber attacks before execution.

Secondary Objectives:

- Analyze temporal network behavior.
- Investigate attack precursor indicators.
- Forecast future cyber attacks.
- Introduce the prediction horizon metric.
- Integrate explainable AI.
- Provide a real-time monitoring dashboard.
- Compare machine learning and deep learning approaches.

## **4. Related Work**

IDS research has evolved through several generations. Signature-based systems such as

Snort focus on known attack patterns. Anomaly detection systems identify deviations from normal behavior. Machine learning approaches introduced automated classification techniques including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, and Gradient Boosting.

Recent advances in deep learning have enabled sequential modeling through Recurrent Neural Networks, Long Short-Term Memory networks (LSTM), and Gated Recurrent Units (GRU). These approaches capture temporal dependencies and have demonstrated promising performance in cybersecurity datasets.

Despite these advances, most studies remain focused on attack detection rather than attack forecasting. Sentinel IDS seeks to address this research gap.

## 5. Proposed Methodology

The proposed pipeline consists of:

Network Traffic Sources

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Traffic Collection

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Feature Extraction

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Flow Generation

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Feature Engineering

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Dataset Preparation

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Machine Learning Models

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Prediction Engine

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Explainable AI

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Dashboard Visualization

Network telemetry is transformed into structured flows. Statistical, protocol-level, and temporal features are extracted. Sequential windows are then generated to create forecasting samples suitable for temporal learning models.

## 6. Datasets

The framework is designed to support multiple public cybersecurity datasets, including:

- CICIDS2017
- CSE-CIC-IDS2018
- UNSW-NB15

These datasets provide realistic attack scenarios and diverse traffic patterns. Final dataset selection, preprocessing strategies, and experimentation remain ongoing. No dataset statistics are reported because the evaluation phase is still under development.

## 7. Machine Learning Models

Traditional Machine Learning Models:

1. Logistic Regression
2. Random Forest
3. XGBoost

These models provide strong baselines, interpretability, and efficient deployment characteristics. XGBoost is particularly attractive because of its performance and ability to handle complex feature interactions.

The study will compare these models using a common preprocessing pipeline.

## 8. Deep Learning Models

Deep learning approaches are expected to play a central role in predictive intrusion detection because attacks often emerge from sequences of events.

Investigated architectures include:

- LSTM
- GRU

Temporal neural networks can capture relationships across time windows and may identify attack preparation phases before malicious actions become observable.

## 9. Explainable Artificial Intelligence

One of the major challenges of AI-driven cybersecurity systems is explainability. Analysts often hesitate to trust black-box predictions without understanding the reasoning process.

Sentinel IDS integrates SHAP (SHapley Additive exPlanations) to:

- Explain individual predictions.
- Rank feature importance.
- Improve transparency.
- Increase analyst trust.
- Support incident investigation.

The explainability layer is designed to convert complex model outputs into actionable intelligence.

## 10. System Architecture

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Python
- Flask

Database:

- SQLite
- PostgreSQL (future scalability)

AI Engine:

- XGBoost
- LSTM
- GRU
- SHAP

Dashboard Components:

- Overview Dashboard
- Network Traffic Analysis
- Threat Intelligence
- Prediction Timeline

- Explainable AI
- Research Metrics
- Dataset Explorer

## 11. Novel Contributions

Contribution 1: Predictive IDS Framework

Contribution 2: Prediction Horizon Metric

Contribution 3: Explainable Forecasting Engine

Contribution 4: Unified Threat Forecasting Platform

Contribution 5: Support for Public and Custom Security Datasets

The prediction horizon metric represents a novel attempt to quantify how early an attack can be forecast before execution.

## 12. Experimental Design

Experiments are currently ongoing.

Planned evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Prediction Horizon

Result tables will be added after experimentation. To maintain scientific integrity, no values are fabricated in this paper.

## 13. Expected Outcomes

Expected outcomes include:

- Earlier threat awareness.
- Improved incident response preparation.
- Increased analyst trust through explainability.
- Enhanced cybersecurity decision support.
- Demonstration of predictive security feasibility.

## 14. Limitations

Potential limitations include class imbalance, dataset bias, concept drift, adversarial behavior, and computational overhead. Predicting novel attacks remains a significant challenge.

## 15. Future Work

Future research directions include:

- Online learning.
- Federated learning.
- Graph neural networks.
- Threat intelligence integration.
- Adaptive forecasting windows.
- Enterprise-scale deployment.

## 16. Conclusion

Sentinel IDS represents an effort to transform intrusion detection from reactive identification toward proactive forecasting. By combining temporal machine learning, deep learning, explainable AI, and real-time analytics, the framework seeks to provide actionable intelligence before attacks occur. Future experimentation will determine the effectiveness and practical deployment feasibility of the proposed approach.

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